

Büşra Tanriverdi, Ph.D.

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PRESENT POSITION

Data Analyst, 2025 – present
Penn Statistics in Imaging and Visualization Center (PennSIVE)
Department of Biostatistics, Epidemiology and Informatics,
Perelman School of Medicine, University of Pennsylvania

EDUCATION

Temple University, Philadelphia, PA, US 2020 – 2024
Ph.D. Psychology
Quantitative Methods Concentration
Dissertation: *Characterizing Threat's Influence in Post-Encoding Reactivation and Its Downstream Effects on Memory Representations*

Koç University, Istanbul, Turkey 2015 – 2017
M.A. Psychological Sciences
Thesis: *The Effect of Interference in Remote versus Recent Autobiographical Memories*

Istanbul Sehir University, Istanbul, Turkey 2010 – 2015
B.A. Psychology

PUBLICATIONS

Hoffman, L., Foley, J., **Tanriverdi, B.**, Chein, J., & Olson, I.R. (2024). Awake Targeted Memory Reactivation Does Not Work. *Memory and Cognition*. DOI: <https://doi.org/10.3758/s13421-024-01576-x>

Tanriverdi, B., Cowan, E.T., Metoki, A., Jobson, K.R., Murty, V.P., Chein, J., & Olson, I.R. (2023). Awake hippocampal-cortical co-reactivation is associated with forgetting. *Journal of Cognitive Neuroscience*. DOI: [10.1162/jocn_a_02021](https://doi.org/10.1162/jocn_a_02021)

Benear, S.L., Popal, H.S., Zheng, Y., **Tanriverdi, B.**, Murty, V.P., Perlman, S.B., Olson, I.R., & Newcombe, N.S. (2023). Setting boundaries: Development of neural and behavioral event cognition in early childhood. *Developmental Science*. DOI: <https://onlinelibrary.wiley.com/doi/10.1111/desc.13409>

Tanriverdi, B., Gregory, D., Olino, T. M., Ely, T. D., Harnett, N., Van Rooij, S. J. H., Lebois, L. A. M., Seligowski, A. V., Javanovic, T., Ressler, K. J., House, S. L., Beaudoin, F. L., Xinming A., Neylan, T. C., Clifford, G. D., Linnstaedt, S. D., Germine, L. T., Bollen, K. A., Rauch, S. L., Haran, J. P., Storrow, A. B., Lewandowski, C., Musey, P. I., Hendry, P. L., Sheikh, S., Jones, C. W., Panches, B. E., Kurz, M. C., McGrath, M. E., Hudak, L. A., Pascual, J. L., Seamon, M. J., Datner, E. M., Pearson, C., Domeier, R. M., Rathlev N. K., O’Neil, B. J., Sanchez, L. D., Bruce, S. E., Miller, M. W., Pietrzak, R. H., Joormann, J., Barch, D. M., Pizzagalli, D. A., Sheridan, J. F., Smoller, J. W., Harte, S. E., Elliott, J. M., McLean, S. A., Kessler, R. C., Koenen, K. C., Stevens, J., & Murty, V. (2022). *Hippocampal Threat Reactivity Interacts with Physiological Arousal to Predict PTSD Symptoms*. *Journal of Neuroscience*, 42 (34) 6593 - 6604. DOI: <https://doi.org/10.1523/JNEUROSCI.0911-21.2022>

Tanriverdi, B., Divleli, A., R., & Celebi, E. (2016). Predictors of support for democratic reform: Role of intergroup friendship, perception of discrimination and identification. In C. Roland-Lévy, P. Denoux, B. Voyer, P. Boski & W. K. Gabrenya Jr. (Eds), *Unity, diversity, and culture: Research and Scholarship Selected from the 22nd Congress of the International Association for Cross-Cultural Psychology* (pp. 231-237).

Manuscripts In Preparation or Under Review:

**denotes equal contribution*

Nguyen, K. V., Skalaban, L. J., **Tanriverdi, B.,** Erardi, J. J., Campos G. A. O., Olson, I. R., Murty V. P., & Newcombe, N. S. (*under review*). Neural Organization of Episodic Memory and Navigation in Children and Adults.

Johnston, C. R., **Tanriverdi, B.,** Sullivan-Toole, H., Reilly, J., Olino, T. M., & Jarcho, J M. (*under review*). Understanding Gaslighting: A Data-Driven Analysis of Themes, Features, and Effects on Subjective Memory. Preprint available: <https://osf.io/preprints/psyarxiv/ypc25>

Tanriverdi, B., Gregory, D. F., Skalaban, L., Olson, I.R., Chein, J., & Murty, V.P. (in preparation). *Characterizing Threat’s Influence on Post-Encoding Consolidation and Its Downstream Effects on Memory Representations*.

Tanriverdi, B., Olson, I.R., Chein, J., & Murty, V.P. (in preparation). *Leveraging Systems Consolidation to Explore Treatment Interventions for Intrusive Memories in PTSD*.

Cowan, E.T.,* **Tanriverdi, B.,*** Murty, V.P., Olson, I.R., & Chein, J. (in preparation). *Encoding repetition leads to hippocampal differentiation*.

Tanriverdi, B.*, Green, H.*, & Chein, J. (in preparation). The Neural and Behavioral Markers of Lower Sensitivity to Social Pressure is Associated with Lower Engagement with Digital Media and Better Mental Well Being.

Gregory, D. F., **Tanriverdi, B.**, Cowan, E., & Murty, V. P. (in preparation). Threat-related arousal bias of amygdala sub-nuclei and the medial temporal lobe during distance estimations in temporal memory of movie clips.

CONFERENCE PRESENTATIONS

**equal contribution*

Tanriverdi, B.*, Green, H.*, Zweben, D., Martinez, S., Skalaban, L., & Chein, J. (October, 2024). *Lower Sensitivity to Social Pressure is Associated with Lower Engagement with Digital Media and Better Mental Well Being*. Accepted for poster presentation at the Society for Neuroscience (SfN), Chicago, IL.

Tanriverdi, B., Skalaban, L., Gregory, D. F., Olson, I. R., Chein, J. & Murty, V. P. (October, 2024). *Early Neural Predictors of Threat-Related Long-Term Memory Reinstatement in Hippocampus*. Accepted for poster presentation at the Society for Neuroscience (SfN), Chicago, IL.

Gregory, D. F., **Tanriverdi, B.**, Cowan, E., Balderston, N. L., & Murty, V. P. (October, 2024). *Threat-related arousal bias of amygdala sub-nuclei and the medial temporal lobe during distance estimations in temporal memory of movie clips*. Accepted for poster presentation at the Society for Neuroscience (SfN), Chicago, IL.

Green, H., **Tanriverdi, B.**, Zweben, D., Martinez, S., Skalaban, L., Chein, J. (September, 2024) *Sensitivity to Social Pressure is Associated with Digital Media Engagement and Better Mental Wellbeing in adults, but Not Adolescents*. Poster presentation at Flux: The Society for Developmental Cognitive Neuroscience 2024, Baltimore, MD.

Nguyen, K.V., **Tanriverdi, B.**, Arantes De Oliveira Campos, G., Olson, I. R., & Newcombe, N.S. (June, 2024). *Neural Encoding of Episodic and Spatial Representations in Development*. Data blitz talk at The Interdisciplinary Navigation Symposium, Merano, Italy.

Tanriverdi, B., Gregory, D.F., Skalaban, L., Olson, I. R., Chein, J., & Murty, V. P. (April, 2024). *Threat-Related Long-Term Memory Reinstatement in Hippocampus*. Poster presentation at The Cognitive Neuroscience Society 31st Annual Meeting, Toronto, Ontario, Canada.

Gregory, D.F., Spangler, B. C., Cowan, E., **Tanriverdi, B.**, & Murty, V. P. (April, 2024). *Horror-evoked Arousal Predicts Biased Distance Estimations for Continuous Events*. Poster presentation at The Cognitive Neuroscience Society 31st Annual Meeting, Toronto, Ontario, Canada.

Tanriverdi, B.*, Cowan, E. T.*, Murty, V. P., Olson, I. R., & Chein, J. (April, 2023). *Encoding Repetition Leads to Hippocampal Differentiation*. Poster presentation at The International Conference on Learning and Memory, Surf City, CA.

- Tanriverdi, B.**, Benear S. L., Metoki, A., Murty, V. P., Chein, J., & Olson, I. R. (April, 2022). *Coordinated hippocampal reactivation during awake rest predicts subsequent forgetting*. Poster presentation at the 29th Annual Meeting of the Cognitive Neuroscience Society (CNS) San Francisco, CA.
- Benear, S.L., Popal, H.S., **Tanriverdi, B.**, Perlman, S.B., Olson, I.R., & Newcombe, N.S. (April, 2022). *Evaluating the neural signatures of event segmentation and memory in children*. Cognitive Neuroscience Society Conference, San Francisco, CA.
- Tanriverdi, B.**, Gregory, D., Ely, T. D., Harnett, N., Van Rooij, S., Lebois, L. A. M., Javanovic, T., Ressler, K. J., Stevens, J., & Murty, V. (November, 2021). *Bilateral hippocampal threat reactivity interacts with physiological arousal to predict PTSD symptoms*. Poster presentation at the 1st Annual Trends in Psychology Summit, virtual hosted by Harvard University Women in Psychology (WiP).
- Johnston, C.R., Sullivan-Toole, H., **Tanriverdi, B.**, Olino, T. M., Jarcho, J. M. (November, 2021). *Was I gaslit? A data-driven, natural language processing-based approach to defining gaslighting*. Poster presentation at the 1st Annual Trends in Psychology Summit, virtual hosted by Harvard University Women in Psychology (WiP).
- Tanriverdi, B.**, Benear S. L., Metoki, A., Murty, V. P., Chein, J., & Olson, I. R. (August, 2021). *Awake coordinated reactivation predicts forgetting*. Poster presentation at the 17th Annual Meeting of the Context and Episodic Memory Symposium (CEMS), Philadelphia, PA.
- Tanriverdi, B.**, Gregory, D., Ely, T. D., Harnett, N., Van Rooij, S., Lebois, L. A. M., Javanovic, T., Ressler, K. J., Stevens, J., & Murty, V. (April, 2021). *Hippocampal threat reactivity interacts with physiological arousal to predict PTSD symptoms*. Poster presentation at the 13th Annual Meeting of the Social and Affective Neuroscience Society (SANS), virtual hosted by SANS.
- Tanriverdi, B.**, & Gülgöz, S. (April, 2017). *The Effect of Interference on Autobiographical Memories*. Poster presentation at 4th International Cognitive Sciences Conference, Hacettepe University, Ankara, Turkey
- Tanriverdi, B.**, Selman, S., B., Pala, S., Duru, A. D., & Bilge, R. (November, 2016). *Tunnel Memory for traumatic events: Is it real?* Poster presentation at Psychonomic Society's 57th Annual Meeting, Boston, MA, USA.
- Bilge, A. R., Selman, S. B., **Tanriverdi, B.** (November, 2016). *Psychology or engineering major, man or woman; It doesn't matter! Spatial representation matters for mental rotation performance*. Poster presentation at Psychonomic Society's 57th Annual Meeting, Boston, MA, USA.
- Tanriverdi, B.**, Divleli, A. R. & Celebi, E. (August, 2014). *Predictors of support for democratic reform: Role of intergroup friendship, perception of discrimination and identification*. Poster

presentation at Congress for Cross-Cultural Psychology (IACCP), 2014, 22nd International, Reims, France.

INVITED TALKS

Guest Speaker, Center for Brain Imaging & Stimulation, Department of Psychiatry, University of Pennsylvania, Philadelphia	May 20th, 2024
Guest Speaker, MiNDS Retreat, Temple University, Philadelphia	Sep 23rd, 2022
Guest Speaker, Dynamic Memory Lab, UC Davis, CA	May 9th, 2022

AWARDS AND HONORS

Dissertation Completion Grant, Temple University	2024
Woodcock Institute Research Grant, Texas Woman's University <i>Role: Co-Investigator (PI: Chein, J. \$15,000)</i>	2023 – 2024
Meta Research PhD Fellowship, META <i>Finalist, Human Computer Interaction - Social Media, People, and Society</i>	2023
Kavli Summer Institute in Neuroscience Fellowship	2022
Statistics Training Award, Temple University <i>For completing Machine Learning Workshop, PITT Methods</i>	Summer, 2021
College of Liberal Arts Summer Research Fellowship, Temple University	Summer, 2021
Outstanding Student Fellowship, Koç University <i>Merit-based award for graduate students</i>	2015 – 2017
Travel award, Koc University <i>For attending Psychonomic Society's 57th Annual Meeting in Boston, MA, US.</i>	2016
Travel and Education award, Jacobs Foundation <i>For attending CCLM Summer School in Weggis, Switzerland.</i>	2016
Travel and Education Award, KU Leuven <i>For attending Perceptual Similarities Summer School</i>	2014
Erasmus + Programme, Turkish National Agency <i>Erasmus traineeship grant for three months</i>	Summer, 2014

Erasmus Programme, Turkish National Agency 2012 – 2013
Erasmus Education grant for nine months

Full scholarships for Undergraduate Studies, Istanbul Sehir University 2010 – 2015
Merit-based award for undergraduate studies

SKILLS

Programming: Bash, HTML, MATLAB, R & RStudio, Python

Software: Acknowledge, FSL, fMRIPrep, MS Office, Adobe Photoshop & Premiere, Psychtoolbox, SPSS,

Languages: Turkish (native), English (fluent)

RESEARCH EXPERIENCE

Graduate Researcher 2020 – 2024
Temple University, Philadelphia, PA, USA.
PIs: Dr. Ingrid Olson, Dr. Jason Chein

Visiting Scholar, Adaptive Memory Lab 2019 – 2020
Temple University, Philadelphia, PA, USA.
PI: Dr. Vishnu P. Murty

Master's Student, Research in Autobiographical Memory Lab 2016 – 2017
Koç University, Istanbul, Turkey.
PI: Dr. Sami Gülgöz

Research Assistant, Memory and Cognition Lab, 2015 – 2016
İstanbul Şehir University, Istanbul, Turkey.
PI: Dr. Reyryan Bilge

Research Assistant, Experimental Psychology Lab, 2014 – 2014
University of Leuven- KU Leuven, Leuven, Belgium.
PI: Dr. Johan Wagemans, Dr. Naoki Kogo, Dr. Nihan Alp

Research Assistant 2013 – 2014
Istanbul Sehir University, Istanbul, Turkey
PI: Dr. Elif Celebi

TEACHING EXPERIENCE

Statistics for Psychology, Lab Instructor Spring, 2024

Department of Psychology, Temple University, Philadelphia, PA.

Brain Matters, TA Fall, 2023

Department of Psychology, Temple University, Philadelphia, PA.

Representational Similarity Analysis (RSA) Workshop Summer, 2022

*Coding Outreach Group, Department of Psychology & Neuroscience,
Temple University, Philadelphia, PA Workshop [link](#)*

Capstone in Neuroscience: Neurodegenerative Diseases, TA Fall, 2021

Department of Psychology, Temple University, Philadelphia, PA.

Intro to Coding Bootcamp-2021, TA Fall, 2021

*Coding Outreach Group, Department of Psychology & Neuroscience,
Temple University, Philadelphia, PA Bootcamp [link](#)*

ADDITIONAL TRAINING & CERTIFICATES

Computational Neuroscience, Neuromatch Academy 2024

Kavli Summer Institute in Cognitive Neuroscience, 2022
Santa Barbara, CA

Applied Machine Learning in R 2021
PITT Methods

Representational Similarity Analysis (RSA) Workshop 2021
Society for Social Neuroscience Pre-Meeting Workshop

Center for Cognition, Learning, and Memory Summer School 2016
University of Bern, Weggis, Switzerland

**Summer School: “Perceptual Organization: Interdisciplinary approaches
and research skills”** 2014
KU Leuven, Leuven, Belgium

LEADERSHIP

Co-founded Interdisciplinary Trauma Discussion Group Temple University, Philadelphia, PA, USA.	2021 – present
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PROFESSIONAL MEMBERSHIPS

Cognitive Neuroscience Society, Member	2020 – present
Social and Affective Neuroscience Society, Member	2019 – present
Psychonomic Society, Member	2016 - present
Society for Neuroscience, Member	2024 - present

AD HOC REVIEWER

Plos ONE

MENTORING

Harry Green, Temple University, Undergraduate Student	2023 - present
Julia Foley, Temple University, Master's Student	2022 - 2024
Katherine Floyd, Temple University, Undergraduate Student	2023 - 2024
Yuliana Fartachuk, Temple University, Undergraduate Student	2022 - 2023
Olivia Bishop, Temple University Undergraduate Student	2022 - 2023
Asude Eracikbas, Bogazici University Graduate	2020 – 2021
Beyza Özen, Koç University Undergraduate Student	2015 – 2017

Popular Media Coverage

The brain's memory hub may explain why some people get PTSD and some don't (August, 2022). In *Inverse* (Article by Miriam Fauzia)

Specific brain responses to traumatic stress linked to PTSD risk (July, 2022). In *Science Daily*

Severity of PTSD symptoms linked to decreased activity in Hippocampus (July, 2022). In *Neuroscience News* (Article by Calli McMurray)

Specific brain responses to traumatic experiences predict vulnerability to develop PTSD
(July, 2022). In *News Medical* (Article by Emily Henderson)